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An Integrated High-Resolution 3D CFD Framework for Viscous Fingering in Porous Media: Relative Permeability, Isotropic Stencils and GPU-Accelerated Solution

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Abstract

Viscous fingering causes channelized displacement in porous media, reducing sweep efficiency and accelerating breakthrough in enhanced oil recovery and geological CO_2 storage. Predicting its three-dimensional morphology remains difficult under high mobility contrast because displacement stability depends simultaneously on saturation-dependent mobility, pressure redistribution, and numerical resolution.

Previous studies have advanced separate aspects of this problem, including fractional-flow and relative-permeability modeling, high-resolution fingering simulation, isotropic discretization, and GPU-accelerated pressure solution. However, these elements have rarely been integrated into one framework for reservoir-relevant 3D fingering analysis.

Here, we develop a high-resolution 3D CFD framework that unifies these components. The model couples Corey-type relative permeability and fractional-flow

phase partitioning with a pressure-based porous-media flow solver. Residual saturation constraints preserve physically meaningful phase depletion limits, while a D3Q27-type stencil and a GPU-accelerated implementation improve directional consistency and computational feasibility.

The framework was first verified against one-dimensional Buckley-Leverett solutions and then validated using Habermann’s quarter five-spot experiments. The simulations reproduced the mobility-ratio-dependent transition from stable displacement to dendritic fingering and captured the decreasing sweep-efficiency trend with increasing mobility ratio, including the contrast between stable and highly unstable regimes.

Three-dimensional simulations revealed tubular channels, peripheral bypass, pressure-field shielding, and volumetric displacement structures that cannot be represented in quasi-two-dimensional models. The stencil-connectivity analysis further showed that controlling grid-aligned propagation is necessary for interpreting 3D morphology. By combining reservoir-relevant constitutive relations with high-resolution CFD, isotropic stencil connectivity, and GPU-based solution, the framework provides a basis for analyzing viscosity-dominated 3D fingering and for extending the model toward complex porous-media displacement.

Keywords: Porous media, Computational fluid dynamics, Viscous fingering, Multiphase, Corey relative permeability

Article Highlights

- A 3D CFD framework integrates Corey relative permeability, fractional flow, isotropic stencils and GPU solution.
- Buckley-Leverett and Habermann benchmarks capture mobility-ratio effects on fingering and sweep efficiency.
- D3Q27-type connectivity reduces grid-aligned growth and reveals tubular channels and pressure shielding in 3D.

1 Introduction

Viscous fingering is a hydrodynamic instability that occurs when a more mobile fluid displaces a less mobile one in a porous medium or a Hele-Shaw type geometry [Saffman and Taylor \(1958\)](#); [Paterson \(1981\)](#). Once the displacement front becomes unstable, the invading phase advances through preferential channels rather than sweeping the domain uniformly. This channelized flow reduces displacement efficiency, accelerates breakthrough, and leaves large volumes of resident fluid bypassed [Buckley and Leverett \(1942\)](#); [Habermann \(1960\)](#). Viscous fingering is therefore a central issue in enhanced oil recovery, groundwater remediation, underground gas storage, and geological carbon dioxide storage, where displacement stability directly affects recovery, injectivity, trapping, and long-term storage security [Gooya et al. \(2019\)](#); [Kang et al. \(2026\)](#); [Wang et al. \(2026\)](#); [Beteta et al. \(2022\)](#).

The fundamental physics of unstable displacement has been studied for more than six decades. Saffman and Taylor established the classical instability mechanism for the penetration of a less viscous fluid into a more viscous fluid in a porous medium or Hele-Shaw cell [Saffman and Taylor \(1958\)](#). Paterson extended this discussion to radial fingering and clarified the preferred wavelength and growth behavior in circular geometries [Paterson \(1981\)](#). In petroleum engineering, Buckley and Leverett provided the foundational one-dimensional theory for immiscible displacement in porous media [Buckley and Leverett \(1942\)](#), while Habermann experimentally demonstrated that miscible displacement efficiency decreases strongly with increasing mobility ratio in quarter five-spot models [Habermann \(1960\)](#). These early studies established two principles that remain essential today: mobility contrast controls displacement stability, and macroscopic sweep efficiency can deteriorate severely even under well-defined injection conditions.

Recent heavy-oil and polymer-flooding studies further demonstrate the practical importance of resolving adverse-mobility-ratio displacement. Skauge et al. used two-dimensional X-ray visualization of Bentheimer slab floods to show that waterfloods at unfavorable mobility ratios develop fingers and channels whose width, growth rate, breakthrough time, and sweep efficiency depend strongly on oil-to-water viscosity ratio [Skauge et al. \(2014\)](#). Their experiments also showed that tertiary polymer injection can mobilize bypassed oil by improving sweep between established water channels and by driving oil-bank formation, emphasizing that the spatial structure left by unstable waterflooding controls later recovery. De Loubens et al. modeled related highly viscous oil waterflood and tertiary polymer-flood experiments and found that conventional Darcy-type simulations with steady-state relative permeability could not reproduce the measured unstable waterflood patterns or breakthrough behavior, even with high-resolution two-dimensional simulations [De Loubens et al. \(2018\)](#). Although adaptive dynamic pore-network models improved the qualitative reproduction of dendritic fingers, their two-dimensional formulation still overestimated breakthrough time and highlighted the need for computationally feasible three-dimensional modeling. These studies are particularly relevant to the present work because they connect adverse mobility ratio, relative permeability, X-ray-observed fingering morphology, and the difficulty of predicting unstable displacement using conventional continuum models.

Subsequent studies showed that viscous fingering cannot be fully understood as a one-dimensional or purely planar phenomenon. Bacri et al. used acoustic measurements to investigate miscible viscous fingering in three-dimensional porous media and demonstrated the transition between diffusive and convective growth regimes [Bacri et al. \(1992\)](#). Zimmerman and Homsy performed early three-dimensional numerical simulations and showed that nonlinear mechanisms such as shielding, spreading, coalescence, and tip splitting observed in two-dimensional studies also appear in three dimensions [Zimmerman and Homsy \(1992\)](#). More recent X-ray computed tomography studies have made this limitation clearer. Suekane et al. directly visualized three-dimensional miscible viscous fingering in packed porous media and quantified the effects of viscosity ratio and Peclet number on sweep efficiency [Suekane et al. \(2017\)](#). Suekane et al. further showed that the gravity number can shift the displacement mode from viscous-dominated fingering to gravitational tongue formation [Suekane et al.](#)

(2019). Yang et al. used micro-focused X-ray CT and three-dimensional printed porous samples to show that low Peclet numbers can produce slender breakthrough fingers, whereas higher Peclet numbers or lower viscosity ratios promote multiple finger structures and enhanced tip splitting Yang et al. (2025). These studies demonstrate the need for three-dimensional analysis, while also showing that complete pressure, saturation, and velocity information is difficult to obtain experimentally inside opaque porous media.

Numerical modeling has therefore become indispensable for interpreting and predicting viscous fingering. Existing numerical approaches, however, emphasize different aspects of the problem, and no single class of methods fully satisfies the requirements of high-resolution, three-dimensional, reservoir-relevant fingering analysis. Reservoir simulators and fractional-flow models naturally incorporate relative permeability, fractional flow, residual saturation, and mobility-ratio dependence, which are essential for engineering interpretations of oil-water, gas-oil, polymer-flood, and CO_2 -water displacement Buckley and Leverett (1942); Bakharev et al. (2020); Khan and Mandal (2021); De Loubens et al. (2018); Beteta et al. (2022). Their strength lies in physical interpretability at the reservoir scale. Their limitation is spatial resolution: unstable fingers are often smeared by numerical dispersion, represented through empirical corrections, or evaluated only through averaged quantities such as breakthrough time and recovery factor. Thus, these models are suitable for field-scale prediction but not for directly resolving the initiation, branching, shielding, and coalescence of individual three-dimensional fingers.

High-order miscible-displacement solvers address a different limitation. Li and Riviere developed high-order discontinuous Galerkin methods for rectilinear, radial, and density-driven miscible fingering and showed that high-order spatial discretization can reduce grid-orientation effects and robustly capture finger growth Li and Rivière (2016). Becker et al. introduced a p-adaptive discontinuous Galerkin method with algebraic multigrid preconditioning, enabling high-resolution three-dimensional miscible fingering simulations at high Peclet number and mobility ratio with reduced computational cost Becker et al. (2018). These studies show that numerical order, mesh resolution, and solver design strongly influence the predicted growth rate and morphology of fingers. However, their formulations are mainly based on miscible displacement with concentration-dependent viscosity. They do not directly provide the saturation-dependent phase mobility, residual saturation treatment, and fractional-flow partitioning required for reservoir-style immiscible or multiphase displacement analysis.

Pore-scale and interface-resolved methods provide the most detailed description of two-phase interfaces. Leclaire et al. developed a generalized three-dimensional lattice Boltzmann color-gradient method for immiscible imbibition and drainage in porous media, providing detailed treatment of multiphase interfaces and wetting behavior Leclaire et al. (2017). Suga et al. developed a D3Q27 multiple-relaxation-time lattice Boltzmann method and demonstrated its stability and accuracy in complex turbulent and porous-medium flows Suga et al. (2015). Lei and Luo used a multiple-relaxation-time lattice Boltzmann model to study miscible viscous fingering with dissolution reactions and showed that fingering onset and late-stage intensity depend on Peclet

number, viscosity ratio, and Damkohler number [Lei and Luo \(2021\)](#). Interface-resolved simulations have also been used for supercritical CO_2 -water displacement. Gooya et al. showed that viscous fingering can dominate supercritical CO_2 -water displacement in fine-grained chalk under geological carbon storage conditions [Gooya et al. \(2019\)](#), while Kang et al. and Wang et al. emphasized the roles of capillary number, viscosity ratio, porosity, and permeability heterogeneity in determining capillary fingering, viscous fingering, saturation, and pressure drop [Kang et al. \(2026\)](#); [Wang et al. \(2026\)](#). These methods are powerful for resolving pore-scale physics, but their computational cost increases rapidly with domain size and geometric complexity. They are also often tied to specific pore geometries, wettability conditions, or CO_2 -water systems, which limits their direct use for larger continuum domains and broad reservoir-style parameter studies.

Core-scale three-dimensional CFD models occupy an intermediate position between detailed pore-scale simulations and field-scale reservoir simulations. Pinilla et al. developed three-dimensional CFD models based on the Navier-Stokes equations and demonstrated that CFD can reproduce common fingering features such as finger initiation, merging, blocking, side branching, and tip splitting [Pinilla et al. \(2021a\)](#). In a related study, Pinilla et al. showed that mesh topology strongly affects predicted fingering patterns in coreflood simulations [Pinilla et al. \(2021b\)](#). Varshney and Pal performed three-dimensional simulations in a cubical domain and analyzed the effects of Reynolds number, Peclet number, and logarithmic mobility ratio on breakthrough and sweep efficiency [Varshney and Pal \(2023\)](#). These studies confirm that three-dimensional CFD can resolve complex morphology that one-dimensional and two-dimensional models cannot capture. At the same time, they reveal a critical numerical challenge: the predicted finger pattern can depend strongly on mesh topology, grid orientation, and discretized flux connectivity between neighboring cells. If this directional bias is not controlled, simulated fingers may partly reflect the numerical grid rather than the physical instability.

Taken together, the comparison above clarifies the remaining methodological gap. Reservoir simulators provide fractional flow and relative permeability but lack resolved three-dimensional fingering morphology. High-order miscible-displacement solvers capture fingering and reduce grid-orientation effects but are not naturally formulated around relative permeability and fractional-flow phase partitioning. Pore-scale and interface-resolved methods can describe detailed interfacial physics but are computationally expensive and difficult to extend to larger continuum domains. Core-scale CFD can reproduce three-dimensional morphology but remains sensitive to mesh-induced anisotropy and often lacks direct coupling with reservoir-style constitutive relations. Relative permeability, fractional flow, CFD, D3Q27-type stencil connectivity, and GPU-based sparse linear solvers have each been used in related contexts [Li and Rivière \(2016\)](#); [Becker et al. \(2018\)](#); [Leclaire et al. \(2017\)](#); [Suga et al. \(2015\)](#); [Skauge et al. \(2014\)](#); [De Loubens et al. \(2018\)](#); [Beteta et al. \(2022\)](#); [Pinilla et al. \(2021a,b\)](#). However, their integrated use within a single framework for resolving three-dimensional fingering with reservoir-relevant saturation-dependent mobility has not been systematically established.

The objective of this study is to develop a high-resolution three-dimensional CFD framework for viscosity-dominated fingering in porous media. The framework aims to bridge two modeling requirements that are often treated separately: reservoir-style saturation-dependent mobility and direct numerical resolution of three-dimensional fingering morphology.

To achieve this, the simulator couples a pressure-based porous-media flow formulation with Corey-type relative permeability and fractional-flow phase partitioning. Residual saturation constraints are introduced to avoid nonphysical phase depletion. A 27-point stencil inspired by the D3Q27 lattice arrangement is used to reduce directional bias in structured-grid pressure and saturation updates, and a GPU-accelerated implementation with a GPU-accelerated AMG-GMRES solver makes the large three-dimensional computations feasible [Becker et al. \(2018\)](#); [Suga et al. \(2015\)](#). In this way, pressure solution, saturation-dependent mobility, phase transport, and isotropic stencil connectivity are integrated into a single mass-conservative computational framework.

The framework is evaluated through a hierarchical validation path. First, one-dimensional Buckley-Leverett displacement solutions are used to verify the implementation of relative permeability and fractional flow [Buckley and Leverett \(1942\)](#). Second, Habermann’s classical quarter five-spot experiments are used to evaluate whether the simulator captures the mobility-ratio dependence of displacement morphology and sweep efficiency [Habermann \(1960\)](#). Third, high-resolution three-dimensional simulations are used to examine volumetric fingering structures, tubular channeling, and pressure-field distortion that cannot be resolved in two-dimensional models [Zimmerman and Homsy \(1992\)](#); [Suekane et al. \(2017\)](#); [Yang et al. \(2025\)](#); [Pinilla et al. \(2021b\)](#); [Varshney and Pal \(2023\)](#). This sequence connects fundamental fractional-flow verification, benchmark fingering validation, and three-dimensional morphological analysis, thereby testing whether the integrated numerical components remain coherent as the physical and geometrical complexity increases.

2 Methodology

2.1 Governing equations

In this study, a Computational Fluid Dynamics (CFD) framework is employed to simulate multiphase flow within porous media. The physical domain is discretized into

Nomenclature

Δp	pressure difference across edge of a bundle[Pa]	$\partial t / \Delta t$	time step[sec]
k	permeability [md]	ϕ	porosity[%]
$\vec{A}$	cross-sectional area of one node’s boundary[m ²]	ρ_m	mixed fluid density[kg/m ³]
ρ_*	density of each phase at standard conditions[kg/m ³]	F	Forchheimer coefficient[-]
C_ϕ	rock compressibility coefficients[Pa ⁻¹]	S_*	saturation of each phase[%]
C_*	non-gaseous phase compressibility coefficient[Pa ⁻¹]	μ_m	mixed fluid viscosity[Pa · s]
C_{me}	effective mass compressibility[kg/(m ³ · Pa)]	V	volume of one node[m ³]
v_m	mixed fluid velocity[m/s]		

cubic control volumes, hereafter referred to as nodes. Fluid flow behavior is simulated based on the state variables within these nodes, which serve as the fundamental computational units. Fluid exchange occurs across the boundary interfaces of these adjacent nodes. For a porous medium model containing N discretization points, the system comprises N corresponding nodes.

In this framework, the pressure drop across the nodes is attributed to both inertial and viscous forces. The governing equation for fluid flow through porous medium is following the Brinkman-Forchheimer form discussed by Nield (1991) for laminar flow. We modified the original equation to model a fluid mixture by incorporating a capillary force term f_c and a gravity term $\rho_m g$, thereby extending its applicability to multiphase systems. The fundamental governing equation is expressed as:

$$\frac{1}{\phi} \frac{\partial(\rho_m \vec{v}_m)}{\partial t} + \frac{1}{\phi^2} \nabla \cdot (\rho_m \vec{v}_m \vec{v}_m) = -\nabla p^{t+1} - \frac{\mu_m}{k} \vec{v}_m + \nabla f_c - \rho_m \vec{g} - \frac{F \rho_m |\vec{v}_m| \vec{v}_m}{k^{1/2}} \quad (1)$$

This study focuses on viscous fingering in laboratory-scale models. Therefore, to isolate the dominant hydrodynamic effects, capillary force f_c , gravity ($\rho_m \vec{g}$) and the Forchheimer inertial drag ($\frac{F \rho_m |\vec{v}_m| \vec{v}_m}{k^{1/2}}$) are neglected in the subsequent formulations throughout the remainder of this paper. Discretizing the base governing equation Equation (1) over time and mixed fluid yields:

$$\frac{1}{\phi} \left(\frac{\rho_m^{t+1} \vec{v}_m^{t+1} - \rho_m^t \vec{v}_m^t}{\Delta t} \right) + H(\vec{v}_m^t) = -\nabla p^{t+1} - \frac{\mu_m}{k} \vec{v}_m^{t+1} \quad (2)$$

$$H(\vec{v}_m^t) = \frac{1}{\phi^2} \nabla \cdot (\rho_m^t \vec{v}_m^t \vec{v}_m^t) \quad (3)$$

where ρ_m^t and $\vec{v}_m^t$ denote the density and velocity of mixture fluid at current time step, while ρ_m^{t+1} and $\vec{v}_m^{t+1}$ represent these properties at the subsequent time step. By combining variables, the flow-rate equation can be further simplified as:

$$\vec{v}_m^{t+1} = \frac{-H(\vec{v}_m^t) - \nabla p^{t+1} + S^t}{a^{t+1}} \quad (4)$$

$$a^{t+1} = \frac{\rho_m^{t+1}}{\phi \Delta t} + \frac{\mu_m}{k} \quad (5)$$

$$S^t = \frac{\rho_m^t \vec{v}_m^t}{\phi \Delta t} \quad (6)$$

This localized calculation model addresses the limitations of traditional macroscopic Darcy-based black-oil models by ensuring mass conservation at microscopic time steps (under 1 second), thereby accurately resolving flow dynamics at the individual control volume scale.

2.2 Implementation of Corey Relative Permeability Model

To partition the total mixture flux into individual phase fluxes, this study combines the standard fractional-flow formulation for two-phase displacement with a Corey-type relative permeability model [Buckley and Leverett \(1942\)](#); [Brooks and Corey \(1964\)](#). In this formulation, endpoint relative permeabilities, residual saturations, and Corey exponents parameterize the saturation dependence of phase mobility. This choice is widely used in reservoir simulation because it preserves a direct link between saturation, mobility ratio, and fractional flow while requiring only a small number of physically interpretable parameters.

Using the fractional flow of water (f_w) in a two-phase water-oil system as an example, the ratio of the water flow rate to the total flow rate is given by:

$$f_w = \frac{\frac{k_{rw}}{\mu_w}}{\frac{k_{rw}}{\mu_w} + \frac{k_{ro}}{\mu_o}} \quad (7)$$

where k_{rw} and k_{ro} are the relative permeabilities of water and oil, respectively. Based on the Corey model, the relative permeabilities are expressed as functions of normalized saturation, adjusted through a transformed saturation parameter, S_i^* . Using water-phase as an example, which S_i^* is defined as:

$$S_w^* = \frac{S_w - S_{wr}}{1 - S_{or} - S_{wr}} \quad (8)$$

Here, S_{wr} and S_{or} represent the residual saturations of water and oil, respectively. Adopting a power-law form, the relative permeabilities are calculated as $k_{rw} = k_{rw}^0 (S_w^*)^{r_w}$ and $k_{ro} = k_{ro}^0 (1 - S_w^*)^{r_o}$. Here, k_{rw}^0 and k_{ro}^0 are the endpoint relative permeabilities, and r_w and r_o are the Corey exponents tuned to reflect the specific multiphase flow characteristics of the porous medium. The normalized saturation S_w^* follows the common effective-saturation treatment used in Corey/Brooks-Corey-type relative permeability formulations, in which residual wetting and non-wetting phase saturations bound the physically accessible saturation interval [Brooks and Corey \(1964\)](#). Substituting these relations into [Equation \(7\)](#) yields the modified fractional flow equation:

$$f_w = \frac{\frac{k_{rw}^0 (S_w^*)^{r_w}}{\mu_w}}{\frac{k_{rw}^0 (S_w^*)^{r_w}}{\mu_w} + \frac{k_{ro}^0 (S_o^*)^{r_o}}{\mu_o}} \quad (9)$$

During the simulation, the residual fluid saturation is strictly monitored at each node. If a phase's saturation drops to or below its residual value, its relative permeability becomes zero, effectively halting its outflow from that specific node. While this section illustrates the method using a two-phase system, the formulation can be extended to n -phase systems in the computational solver.

2.3 Solution for multi-phase flow

The formulation is based on the following physical assumptions: (1) fluid flow occurs exclusively across the boundaries of the control volumes; (2) the fluids are continuous and exhibit Newtonian behavior; (3) mass and energy transfer between the fluids and the rock matrix are negligible; and (4) both the fluids and the rock matrix possess a defined degree of compressibility.

The density (ρ_m) and viscosity (μ_m) of the fluid mixture are determined using a saturation-weighted averaging approach based on Corey exponents:

$$\mu_m = \left[\sum_{i=1}^n (k_{ri}^0 S_i \mu_i^{-r_i}) \right]^{-1/r_m} \quad (10)$$

$$\rho_m = \left[\sum_{i=1}^n (k_{ri}^0 S_i \rho_i^{-r_i}) \right]^{-1/r_m} \quad (11)$$

where i denotes the phase index, S_i is the saturation of phase i in a given node, and r_m is the mean Corey exponent over all phases, calculated as $\frac{1}{n} \sum_{i=1}^n r_i$.

Specifically, the mixture viscosity and density are treated as effective properties for the pressure equation and are evaluated using an empirical saturation-weighted power-mean approximation. This approximation is not used to define phase partitioning; phase fluxes are determined by the fractional-flow formulation and Corey-type relative permeabilities. In the validation cases presented here, fixed Corey exponents are used for all phases, which preserves dimensional consistency of the effective-property calculation.

Due to the compressibility of both the rock and the fluids, pressure transients propagate through the medium. The total compressibility effect is evaluated using the continuity equation:

$$\frac{\partial(\phi \rho_m)}{\partial t} = \phi \frac{\partial \rho_m}{\partial t} + \rho_m C_\phi \phi \frac{\partial p}{\partial t} \quad (12)$$

The numerical simulation of multi-phase flow is based on solving the mass balance equation derived from Navier-Stokes and Brinkman-Forchheimer system. Using an example of 3-phase (oil, gas and water) fluid simulation, upon separating the variables from [Equations \(10\) and \(11\)](#), the equation would be as follows:

$$\frac{\partial(\phi \rho_m)}{\partial t} = \phi (S_o C_o \rho_o + S_g C_g \rho_g + S_w C_w \rho_w + \rho_m C_\phi) \frac{\partial p}{\partial t} = C_{me} \frac{\partial p}{\partial t} \quad (13)$$

Where C_{me} denotes the effective mass compressibility of the multiphase mixture in the porous medium. This term couples the phase saturations, individual component compressibilities, and rock porosity into a single coefficient for the pressure time-derivative. This relationship allows the pressure formulation to be expressed as:

$$\frac{\partial(\phi \rho_m)}{\partial t} = C_{me} \frac{\partial p}{\partial t} \approx C_{me} \frac{p^{t+1} - p^t}{\Delta t} \quad (14)$$

For the gaseous phase (if present), the isothermal compressibility c_g is derived dynamically using the Z-factor, computed via [Kareem et al. \(2016\)](#) correlation:

$$z = \frac{0.27P_{pr}}{yT_{pr}} \quad (15)$$

Where z is the gas compressibility factor (z -factor), P_{pr} is Pseudo-reduced pressure, and T_{pr} means Pseudo-reduced temperature. y is the root of following equation based on empirical saturation-weighted mixing rule:

$$\left[R_5 y^2 (1 + A_{11} y^2) e^{(-A_{11} y^2)} \right] + R_1 y - \frac{R_2}{y} + R_3 y^2 - R_4 y^5 + 1 = 0 \quad (16)$$

The coefficients in this equation are defined as follows:

$$\begin{aligned} R_1 &= A_1 + \frac{A_2}{T_{pr}} + \frac{A_3}{T_{pr}^3} + \frac{A_4}{T_{pr}^4} + \frac{A_5}{T_{pr}^5}, & R_2 &= \frac{0.27P_{pr}}{T_{pr}}, \\ R_3 &= A_6 + \frac{A_7}{T_{pr}} + \frac{A_8}{T_{pr}^2}, & R_4 &= A_9 \left(\frac{A_7}{T_{pr}} + \frac{A_8}{T_{pr}^2} \right), & R_5 &= \frac{A_{10}}{T_{pr}^3} \end{aligned}$$

And the static parameters A_n is defined as:

$$\begin{aligned} A_1 &= 0.3265, & A_2 &= -1.07, & A_3 &= -0.5339, & A_4 &= 0.01569, & A_5 &= -0.05169, \\ A_6 &= 0.5475, & A_7 &= 0.7361, & A_8 &= 0.1844, & A_9 &= 0.1056, & A_{10} &= 0.6134, & A_{11} &= 0.721 \end{aligned}$$

After the z -factor is obtained, gas compressibility can be calculated as:

$$C_g = \frac{1}{P} - \frac{1}{z} \left(\frac{\partial z}{\partial P_z} \right)_{T_z} \quad (17)$$

where P_z and T_z represent absolute pressure and temperature at which the z -factor was calculated.

2.4 Momentum Equation and Pressure Solver

To advance the flow calculation to the next time step, the pressure field must be updated simultaneously for all nodes. The mass balance is enforced by the integral form of the general transport equation:

$$\frac{\partial \phi \rho_m}{\partial t} + \nabla \cdot (\rho_m \vec{v}_m) = 0 \quad (18)$$

Substituting the derived velocity field [Equation \(4\)](#) into the divergence theorem form yields:

$$\begin{aligned}\nabla \cdot (\overrightarrow{v_m^{t+1}}) &= \nabla \cdot \left[\frac{1}{a^{t+1}} \left(-H(\overrightarrow{v_m^t}) - \nabla p^{t+1} + S^t \right) \right] \\ &= \nabla \cdot \left[\frac{1}{a^{t+1}} (-H(\overrightarrow{v_m^t}) + S^t) \right] - \nabla \cdot \left(\frac{\nabla p^{t+1}}{a^{t+1}} \right)\end{aligned}\quad (19)$$

Combining this with [Equation \(14\)](#), the implicit pressure equation is formulated as:

$$\begin{aligned}\frac{\partial(\phi \rho_m)}{\partial t} + \nabla \cdot (\overrightarrow{v_m^t}) &= \\ C_{me} \frac{p^{t+1} - p^t}{\Delta t} + \nabla \cdot \left(\frac{-H(\overrightarrow{v_m^t}) + S^t}{a^{t+1}} \right) - \nabla \cdot \left(\frac{\nabla p^{t+1}}{a^{t+1}} \right) &= 0\end{aligned}\quad (20)$$

By discretizing and iterating this equation over all computational nodes, a linear system of equations is constructed to solve for the unknown pressure field at the next time step (p^{t+1}). This system is structured in the standard matrix format $[A][X]^{t+\Delta t} = [B]^t$, where the coefficients are defined as:

$$[A]^{t+\Delta t} = \begin{cases} \left(\frac{1}{a^{t+1}} \right)_{ij} & \text{if } i \neq j, \quad i = 1, 2, 3, \dots, N \\ \frac{C_{me}}{\Delta t} + \sum_{j=1}^N \left(\frac{1}{a^{t+1}} \right)_{ij} & \text{if } i = j, \quad j = 1, 2, 3, \dots, N \end{cases} \quad (21)$$

$$[B]^t = \left[C_{me} \frac{p^t}{\Delta t} - \nabla \cdot \left[\frac{1}{a^{t+1}} \left(-H(\overrightarrow{v_m^t}) + S^t \right) \right] \right]_i \quad i = 1, 2, 3, \dots, N \quad (22)$$

$$[X]^{t+\Delta t} = [p^{t+1}]_i \quad i = 1, 2, 3, \dots, N \quad (23)$$

In these formulations, $[A]$ is an $N \times N$ sparse coefficient matrix representing the hydraulic conductivity between nodes. $[X]^{t+\Delta t}$ is the solution vector containing the updated pressure field, and $[B]^t$ is the explicit source vector derived from the previous time step's state variables and boundary conditions.

Because the evolving fluid interfaces can lead to highly ill-conditioned matrices, this system is solved robustly using the Algebraic Multi-grid Generalized Minimal Residual (AMG-GMRES) algorithm provided by the CUSP library [Olson \(2025\)](#), accelerated via GPU to ensure computational feasibility.

2.5 Multi-phase Saturation Update

Once the new pressure field and flow rates have been obtained, the fluid saturation in each node is updated explicitly. The stable time step Δt is strictly governed by the algorithm-imposed Courant-Friedrichs-Lewy (CFL) condition. Taking the water phase as an example, once the pressure field is obtained, the water-phase velocity can

be calculated as $v_w = f_w v_m$. Accordingly, the saturation update for internal nodes is given by:

$$S_w^t = \Delta t \frac{\nabla \cdot v_w A}{l^3 \phi} + \frac{S_w^{t-1} \rho_w^{t-1}}{\rho_w^t} \quad (24)$$

For boundary nodes handling fluid injection and production, the update equations are modified respectively:

$$S_w^t = \Delta t \frac{A(\nabla \cdot v_w + v_{w(in)})}{l^3 \phi} + \frac{S_w^{t-1} \rho_w^{t-1}}{\rho_w^t} \quad (25)$$

$$S_w^t = \Delta t \frac{A(\nabla \cdot v_w - v_{w(out)})}{l^3 \phi} + \frac{S_w^{t-1} \rho_w^{t-1}}{\rho_w^t} \quad (26)$$

where $v_{w(in)}$ and $v_{w(out)}$ represent the boundary mass flow rates, and l represents the cubic edge length of a control volume (m). This time-stepping and flow procedure proceeds until the target injection volume or a predefined saturation threshold is achieved.

2.6 D3Q27 Isotropic Discretization Stencil

In simulating viscous fingering within a grid-based CFD framework, the choice of discretization stencil plays a critical role in the morphological accuracy of the displacement front.

As demonstrated by [Pinilla et al. \(2021b\)](#), mesh topology heavily influences the predicted shape of viscous fingers. Standard finite volume methods typically employ a 7-point stencil (flux across 6 faces) for 3D structured grids. However, it is well documented that such low-order connectivity can introduce significant grid orientation effects, artificially promoting fluid flow along the grid axes and suppressing diagonal growth.

To reduce this numerical artifact and improve the directional uniformity of pressure and saturation propagation, this study adopts a 27-point stencil inspired by the D3Q27 lattice arrangement used in the Lattice Boltzmann Method (LBM). Although the governing equations are solved using a Navier-Stokes/Brinkman-Forchheimer-based porous-media flow formulation rather than a full LBM solver, the neighboring-cell flux weights follow the D3Q27 face-, edge-, and corner-connectivity pattern. The stencil therefore incorporates 26 neighboring nodes (6 faces, 12 edges, and 8 corners), providing a more isotropic local coupling than the conventional 7-point stencil on structured cubic grids.

By incorporating these geometric weights into the multiphase flux calculation, the solver accounts for multidirectional flow contributions and reduces grid-orientation constraints. The purpose is not to eliminate numerical anisotropy completely, but to suppress premature grid-aligned growth so that the resolved fingering morphology is less dominated by structured-grid connectivity.

As illustrated in [Fig. 1](#), the discrete velocity directions (c_i) and their corresponding weights (w_i) are defined as:

$$c_i = \begin{cases} (0, 0, 0) & i = 0 \\ (\pm l, 0, 0), (0, \pm l, 0), (0, 0, \pm l) & i = 1 \sim 6 \\ (\pm l, \pm l, 0), (\pm l, 0, \pm l), (0, \pm l, \pm l) & i = 7 \sim 18 \\ (\pm l, \pm l, \pm l) & i = 19 \sim 26 \end{cases} \quad (27)$$

The discrete velocity set is defined as follows:

$$w_i = \begin{cases} w_{center} = \frac{8}{27} & i = 0 \\ w_{face} = \frac{2}{27} & i = 1 \sim 6 \\ w_{edge} = \frac{1}{54} & i = 7 \sim 18 \\ w_{corner} = \frac{1}{216} & i = 19 \sim 26 \end{cases} \quad (28)$$

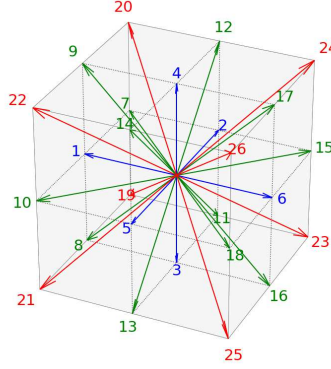


Fig. 1: D3Q27 discretization stencil model

In the present finite-volume implementation, the D3Q27-inspired weights are used as directional coupling coefficients for neighboring-cell pressure and saturation exchange. Face, edge, and corner connections are included in the local stencil, and the weighted contributions are normalized over the available neighbors so that the total outgoing flux from each control volume remains conservative. The stencil is therefore used as an isotropic connectivity operator within the pressure and saturation updates, not as a full lattice Boltzmann collision-streaming scheme.

Because the number and orientation of connected neighboring cells directly affect the numerical propagation of pressure and saturation, the influence of stencil connectivity was examined explicitly. The conventional 7-point stencil connects each cell only to its six face-neighboring cells, whereas the D3Q27-type stencil connects each cell to

26 surrounding neighbors through face, edge, and corner directions. The comparison between these two stencil connectivities is presented as a numerical sensitivity analysis in [Section 3.3](#) and [Section 4](#).

3 Numerical Verification and Experimental Validation

Verification of the simulator’s ability to reproduce the target fractional-flow behavior and benchmark fingering trends is a crucial step in model development. This section presents a two-stage validation: first, a fundamental 1D verification against the Buckley-Leverett analytical solution, and second, a quasi-2D benchmark comparison against the classic viscous fingering experiments conducted by [Habermann \(1960\)](#). For consistency, all simulation cases in this study employ fixed Corey exponents of $r_w = r_o = 2$ and endpoint relative permeabilities of $k_{ro} = k_{rw} = 1$. The corresponding relative permeability and fractional-flow curves are presented in [Fig. 2](#). Additionally, the mobility ratio is defined based on fluid viscosities as $M = \mu_{displaced}/\mu_{displacing}$, with the displacing phase (water) viscosity fixed at $0.001 \text{ Pa} \cdot \text{s}$. Moreover, the CFL number used in all simulations in this study is fixed at 0.1.

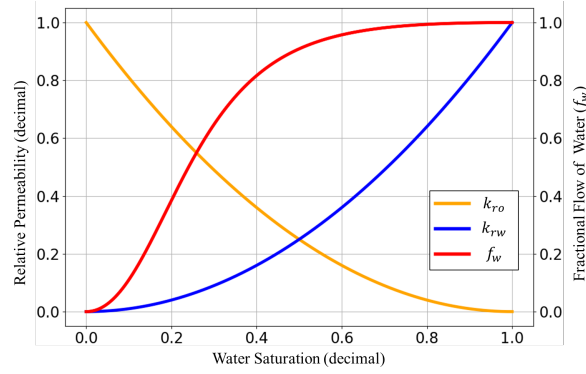


Fig. 2: Relative permeability and fractional-flow curves used in this study

3.1 One-Dimensional Verification (Buckley-Leverett)

To validate the fundamental transport equations and the implementation of the Corey relative permeability model described in [Section 2.2](#), simplified 1D two-phase flow scenarios were simulated. Two fluid types (water and glycerol) were considered in this verification. The fluid properties and model dimensions are summarized in [Table 1](#). The verification cases utilized a completely homogeneous domain, initially fully saturated with either glycerol or water as non-wetting phase. The simulator injected the displacing fluid at a constant flow rate of $10^{-10} \text{ m}^3/\text{s}$ with glycerol or water as wetting phase.

To verify the relative permeability model, a comparison between the numerical results and the semi-analytical solution presented by [Leverett \(1941\)](#) is shown in [Fig. 3](#).

Table 1: 1D model conditions used in this study

Variable	Value	Variable	Value
Node length(m)	0.000762	Model size(nodes)	$200 \times 1 \times 1$
Porosity(%)	0.26		$152.4 \text{ mm} \times 0.762 \text{ mm} \times 0.762 \text{ mm}$
Permeability(md)	1000	Water density(kg/m ³)	1000
Initial pressure(MPa)	0.1	Glycerol density(kg/m ³)	800
$S_{or}, S_{wc}(\%)$	0	Water viscosity(Pa · s)	0.001
		Glycerol viscosity(Pa · s)	0.01

Fig. 3 presents the simulation and semi-analytical results for Case A (A1 and A2): water injection into a glycerol-saturated reservoir, and Case B (B1 and B2): glycerol injection into a water-saturated reservoir. Subfigures A1 and B1 show visual representations of the water and glycerol saturation profiles over time. Subfigures A2 and B2 plot the semi-analytical (dots) and numerical (lines) solutions based on the Brooks-Corey relative permeability models.

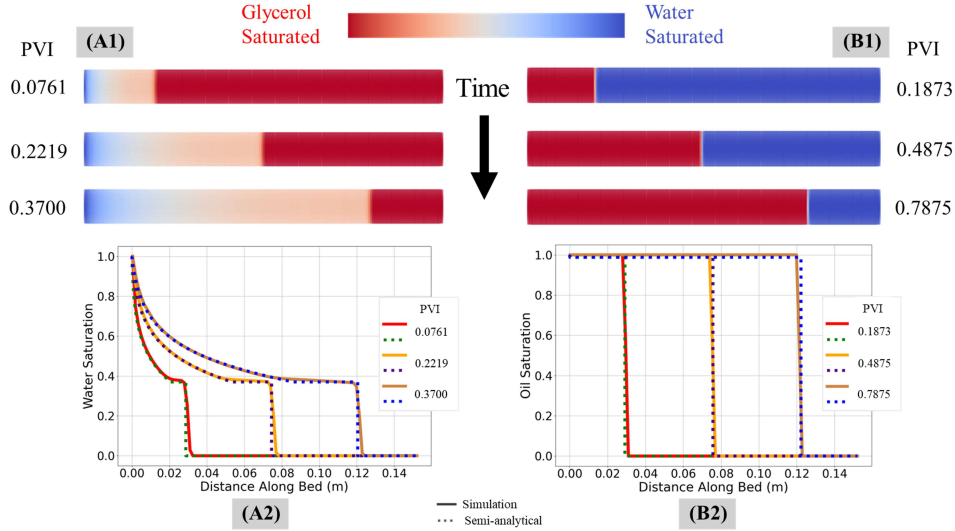


Fig. 3: Comparison of the pore-volume-injected dependent saturation profiles calculated by the proposed simulator and the Buckley-Leverett semi-analytical solution

In Case A, as water displaces glycerol, it creates a saturation profile characterized by a sharp water shock at the front, followed by a gradual saturation gradient. In Case B, because the injected glycerol has a lower mobility than the defending water,

the displacement is highly stable, resulting in a piston-like displacement with virtually no saturation gradient behind the fluid front. Overall, good agreement is observed between our numerical simulations and the semi-analytical solutions, confirming the solver’s accuracy.

3.2 Two-Dimensional benchmark comparison against Physical Experiments

Following the fundamental 1D verification, the simulator was further evaluated through a validation-oriented comparison with Habermann’s classic quarter five-spot experiments [Habermann \(1960\)](#). This section compares the simulation results against Habermann’s ”quarter of a five-spot” experiments, focusing on displacement patterns and sweep efficiency under various mobility ratios.

Habermann’s experiments were miscible displacement experiments, whereas the present simulator uses an effective saturation-based fractional-flow formulation with Corey-type mobility functions. Therefore, the comparison is not intended to reproduce pore-scale miscibility or molecular dispersion directly. Instead, Habermann’s data are used as a benchmark for testing whether the simulator captures the macroscopic mobility-ratio dependence of displacement morphology and sweep efficiency.

3.2.1 Simulation Setup

The simulation domain was configured to replicate the geometry of Habermann’s artificially consolidated sand models, representing a quarter of a five-spot pattern. This configuration consists of a square reservoir section with an injection well located at one corner and a production well at the diagonally opposite corner. To ensure physical consistency with the experimental benchmarks—specifically Models A and C from Habermann’s work—the simulation parameters were initialized using the values listed in [Table 2](#). Fluid was injected at a constant injection rate of $4.2 \times 10^{-7} \text{ m}^3/\text{s}$, while a constant pressure boundary condition was maintained at the outlet. This study examined six distinct mobility ratios (M), ranging from a highly stable ratio of 0.151 to a highly unstable ratio of 71.5, identical to the experimental conditions reported by [Habermann \(1960\)](#).

Table 2: Initial conditions in quasi-2D simulations

Variable	Value	Variable	Value
Node length(m)	0.000762	Model size(nodes)	$500 \times 500 \times 3$
Porosity(%)	0.26		$381 \text{ mm} \times 381 \text{ mm} \times 2.286 \text{ mm}$
Permeability(md)	11843.08	Density water(kg/m^3)	1000
Initial pressure(MPa)	0.1	Density oil(kg/m^3)	800
S_{or}, S_{wc} (%)	10		

3.2.2 Qualitative Comparison of Displacement Fronts

The morphological evolution of the displacement front is compared in Fig. 4, where the top row (A^* through F^*) displays the concentration maps generated by the simulator, and the bottom row (A through F) shows the corresponding sketches from Habermann’s experiments. Both sets of results correspond to the same Pore Volume Injected (PVI) at the moment of breakthrough or at specific intermediate stages.

In the stable displacement regime ($M \leq 1$), the simulator exhibits good agreement with experimental observations. For the most stable case ($M = 0.151$, Case A^*), the displacement front maintains a convex, piston-like shape that efficiently sweeps the domain. At a unity mobility ratio ($M = 1.0$, Case B^*), the front develops the characteristic cusp shape near the production well, driven by the potential pressure gradient inherent to the five-spot geometry. This confirms that the solver correctly resolves the underlying macroscopic potential flow field.

In the unstable regime ($M > 1$), the transition to viscous fingering is distinctly captured. As the mobility ratio increases to 2.40 (Case C^*), initial perturbations emerge on the front, mirroring the experimental onset of instability. For higher ratios ($M = 4.58$ to 17.3), the simulator reproduces the "shielding effect", where larger, dominant fingers advance rapidly, suppressing the growth of adjacent smaller fingers. Notably, in Case E^* ($M = 17.3$), the simulation generates a branching structure that closely resembles the dendritic patterns observed in the experiment (E). At the most extreme condition ($M = 71.5$, Case F^*), the flow forms a highly complex, tree-like network with fine branching.

It is important to note that, although the simulation captures the main morphological trends, including shielding, branching density, and overall pattern complexity, the exact topological configuration of individual fingers differs between the simulation and experiments. This discrepancy is physically inherent to the stochastic nature of hydrodynamic instability. Viscous fingering is driven by microscopic perturbations at the interface. In Habermann’s physical experiments, these perturbations arise from inevitable pore-scale heterogeneities within the consolidated sand packs Habermann (1960). In contrast, the current ideal simulation is triggered by minimal numerical noise or grid orientation. Consequently, because the specific locations of finger initiation are governed by random perturbations, the exact spatial pattern cannot be deterministically matched. However, the simulator successfully captures the macroscopic flow regime and the structural complexity characteristic of each specific mobility ratio.

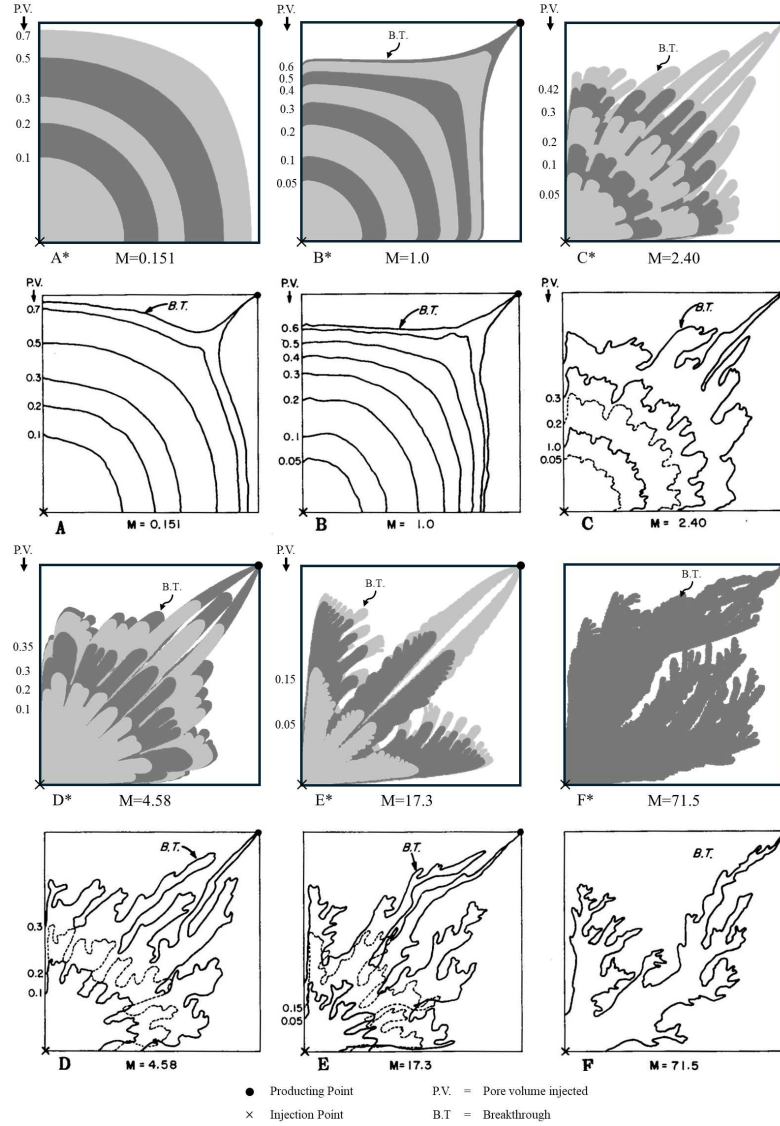


Fig. 4: Comparison of displacement fronts in a quarter five-spot pattern.
Row 1 and 3 (A*-F*): Simulation results of water-saturation distribution.
Row 2 and 4 (A – F): Experimental sketches adapted from [Habermann \(1960\)](#).
 Left to right corresponds to increasing mobility ratios (M) from 0.151 to 71.5

3.2.3 Quantitative Analysis of Sweep Efficiency

To provide a quantitative comparison beyond visual comparison, the sweep efficiency (recovery rate) at the moment of breakthrough was analyzed. [Fig. 5](#) plots the sweep

efficiency against the mobility ratio (M) on a logarithmic scale, comparing the simulation results with Habermann's experimental data. In this study, breakthrough is defined as the time at which the injected fluid first reaches the production well, and a snapshot of the state at all nodes across the model is taken at that time. The analysis reveals that the simulator correctly reproduces the exponential decay trend of sweep efficiency as the mobility ratio increases. However, a detailed comparison highlights a "sensitivity crossover" between the numerical and experimental results. In the stable regime ($M = 0.151$), the simulation predicts a sweep efficiency of 90%, which is higher than the experimental value of 76%. This suggests that the numerical model, being ideally homogeneous, produces a near-perfect piston-like displacement, whereas physical experiments inherently contain microscopic heterogeneities that slightly reduce efficiency. At a unity mobility ratio ($M \approx 1$), the simulation yields a sweep efficiency of 63%, showing reasonable agreement with the experimental value of 67%. This proximity validates the accuracy of the base flow solver in the absence of strong viscosity contrasts.

Conversely, in the highly unstable regime ($M > 1$), the simulation predicts lower sweep efficiency compared to the experimental data (e.g., 13% vs. 21% at $M = 71.5$). This indicates that the simulated primary fingers advance more rapidly toward the outlet. This lower sweep efficiency with increasing M is attributed to numerical idealizations that affect flow dynamics. First, the model lacks physical transverse dispersion inherent to physical sand packs. In experiments, physical dispersion tends to broaden finger tips and retard their longitudinal growth velocity; in the simulation, fingers remain sharp and advance rapidly.

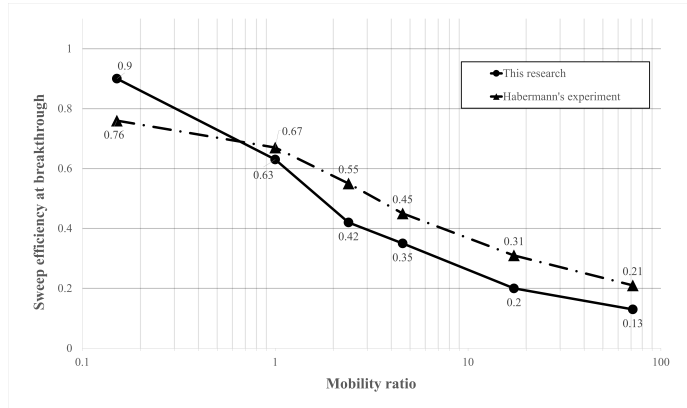


Fig. 5: Sweep efficiency at breakthrough as a function of mobility ratio

Solid line : Simulation results from this study

Dotted line: Characteristic behavior digitized by [Habermann \(1960\)](#) at Fig.3

Second, residual grid orientation effects inherent to the five-spot diagonal configuration may slightly accelerate diagonal channeling. Despite these idealized deviations,

the results confirm that the developed simulator reproduces the primary regime transition and captures the critical dependence of displacement efficiency on mobility ratio.

Breakthrough was defined as the first time at which the injected phase reached the production boundary above the prescribed saturation threshold. The experimental sweep-efficiency values were extracted from Habermann’s reported breakthrough recovery data, while the simulated values were computed as the injected-phase occupied pore saturation at breakthrough in the whole model.

3.3 Numerical Sensitivity to Stencil Connectivity

The benchmark comparison against Habermann’s five-spot experiments supports that the simulator captures the mobility-ratio dependence of displacement morphology and sweep efficiency. However, the original experimental data do not provide modern image-derived quantities such as invaded volume, interfacial area, or connected-cluster statistics. These quantities are therefore not used here as experimental validation metrics. Instead, this subsection examines whether the computed fingering pattern is sensitive to stencil connectivity.

The conventional 7-point stencil and the proposed D3Q27-type stencil were compared under the same quasi-two-dimensional five-spot conditions. The comparison was performed for three unstable mobility ratios, $M = 4.58$, 17.3 , and 71.5 . The 7-point stencil connects each cell only through face-neighboring cells, whereas the D3Q27-type stencil includes face, edge, and corner connections. This difference directly affects how pressure and saturation disturbances propagate through the structured grid.

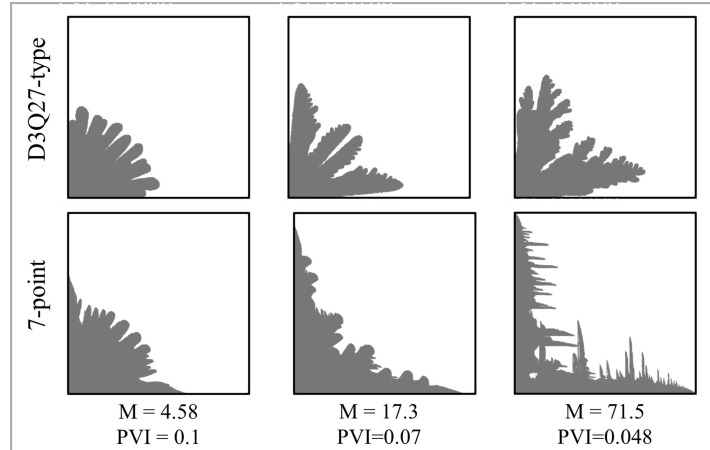


Fig. 6: Effect of stencil connectivity on quasi-two-dimensional fingering patterns for $M = 4.58$, 17.3 , and 71.5

Fig. 6 shows that the 7-point stencil tends to promote grid-aligned channeling, especially at high mobility ratio. In contrast, the D3Q27-type stencil distributes the

displacement over a wider range of directions, producing a more spatially dispersed fingering pattern and reducing preferential grid-aligned propagation compared with the conventional 7-point stencil.

Table 3: Stencil-connectivity comparison

Mobility ratio (M)	Stencil	Breakthrough PVI ¹	Runtime per iteration ²
4.58	7-point	0.30	142.29 ms
	D3Q27-type	0.34	156.04 ms
17.3	7-point	0.21	83.13 ms
	D3Q27-type	0.20	199.34 ms
71.5	7-point	0.10	88.50 ms
	D3Q27-type	0.13	270.15 ms

¹ Snapshots are shown at selected pre-breakthrough PVI values shown in Fig. 6.

² Runtime per iteration = total computation time / number of iterations.

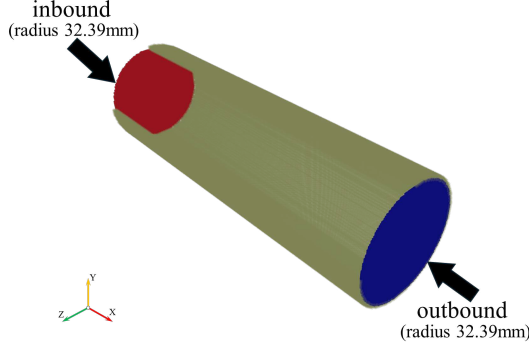
Additionally, the improved directional behavior is accompanied by an increase in per-iteration compute cost and a slight change in breakthrough PVI, as shown in Table 3. These differences are expected because the D3Q27-type stencil introduces broader local connectivity. Nevertheless, these additional effects remain small in the two-dimensional cases, whereas the reduction in grid-oriented propagation is important for interpreting fingering at high mobility ratios. However, the breakthrough PVI results also show that the D3Q27-type stencil primarily reduces directional bias and does not necessarily improve breakthrough PVI.

4 Three-Dimensional Simulation

While the primary validation of this study focused on 1D and quasi-2D systems where experimental benchmarks are available, the developed CFD framework is inherently designed for three-dimensional analysis. To demonstrate this capability, a large-scale 3D simulation was conducted as a preliminary investigation into volumetric fingering dynamics. The 3D model consists of a domain discretized into $290 \times 91 \times 91$ nodes (approximately 2.4 million control volumes), employing a cylindrical injection scheme. Apart from the model dimensions and boundary conditions (illustrated in Fig. 7), the physical parameters were maintained identical to those listed in Table 4. This study examined two distinct mobility ratios (M) at an unstable ratio of 17.3 and a highly unstable ratio of 71.5.

Table 4: Initial conditions for the 3D simulations

Variable	Value	Variable	Value
Node length (m)	0.000762	Model size(nodes)	$290 \times 91 \times 91$
Porosity(%)	0.26		$221.7 \text{ mm} \times 69.3 \text{ mm} \times 69.3 \text{ mm}$
Permeability (md)	11843.08	Density water (kg/m^3)	1000
Initial pressure (MPa)	0.1	Density oil (kg/m^3)	800
$S_{or}, S_{wc}(\%)$	10		

**Fig. 7:** Boundary condition of the 3D model

Before interpreting the three-dimensional fingering morphology, the influence of stencil connectivity was examined in the 3D model for the most unstable case, $M = 71.5$, using snapshots at $\text{PVI} = 0.038$. As shown in Fig. 8, the conventional 7-point stencil produced elongated axial channels that propagated rapidly toward the outlet, indicating a directional preference associated with face-neighbor connectivity. In contrast, the D3Q27-type stencil produced broader radial spreading and a more volumetric displacement pattern. The breakthrough PVI increased from 0.05 with the 7-point stencil to 0.22 with the D3Q27-type stencil, while the runtime per iteration increased from 354.49 ms to 766.90 ms. Although the quasi-two-dimensional five-spot tests and the three-dimensional cylindrical model are not geometrically identical, this result shows that stencil choice can strongly affect the predicted volumetric fingering pattern under highly unstable conditions. Therefore, the following three-dimensional analysis uses the D3Q27-type stencil to reduce premature grid-aligned breakthrough and to focus on volumetric fingering behavior.

Fig. 9 presents the simulation results for the representative unstable cases ($M = 17.3$ and 71.5). Unlike quasi-2D approximations, the 3D results reveal the complex spatial connectivity of the fingers. The fluid channels are not merely planar branches but form intricate, tubular structures that can intertwine and bypass oil in the depth direction—a degree of freedom physically unavailable in quasi-2D models. These tubular

structures emerge because the displacing fluid in the peripheral regions advances significantly faster than in the central core. This behavior is driven by a three-dimensional "shielding effect", where pressure interference is maximized at the center.

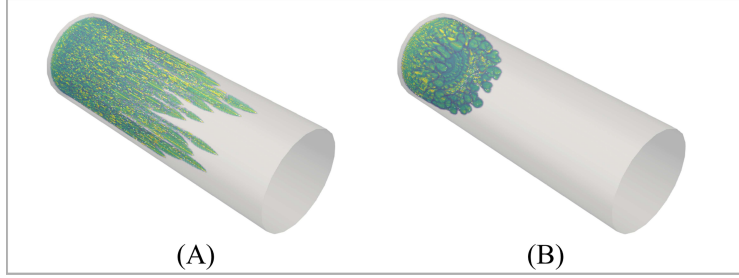


Fig. 8: Three-dimensional comparison of fingering morphology at $M = 71.5$ and $PVI = 0.038$: (A) conventional 7-point stencil and (B) proposed D3Q27-type stencil

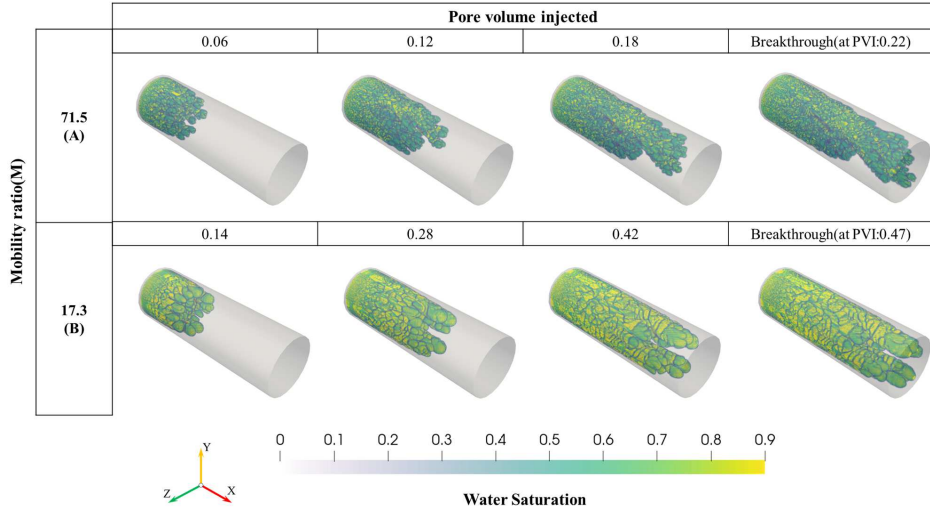


Fig. 9: Water-saturation evolution in the 3D model with $M = 71.5$ (A) and 17.3 (B)

To elucidate the "shielding effect" observed at high mobility ratios, Fig. 10 presents a 2D cross-sectional slice of the 3D pressure field for the highly unstable case ($M = 71.5$) at $PVI = 0.06$. The pressure distribution suggests that finger growth is closely associated with the dynamic redistribution of pressure around high-mobility displacement pathways. As the less viscous fluid penetrates the medium, the high-pressure region (indicated by the red and yellow contours) propagates along the primary fingers because of their higher mobility while reducing the driving force available to adjacent

regions. The D3Q27-type stencil helps represent this radial bypass behavior by reducing artificial axial confinement and allowing flux exchange through face, edge, and corner connections.

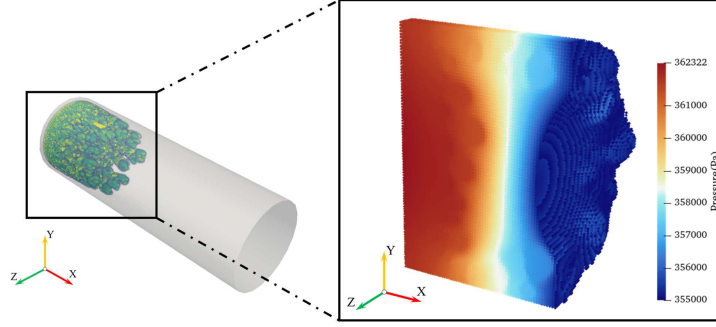


Fig. 10: Cross-sectional slice of the 3D pressure corresponding to Fig. 9(A) at $PVI = 0.06$

Because similar phenomena have been observed in Suekane’s X-ray CT experiments [Suekane et al. \(2017\)](#) and Pinilla’s numerical studies [Pinilla et al. \(2021b\)](#), the results of this study qualitatively support the framework’s capability to handle high-resolution 3D grids and capture the volumetric nature of viscous instability. However, more quantitative validation against 3D experimental data will be required in future work.

Additionally, this study utilized a standard workstation equipped with an 8-core AMD Ryzen 7700 CPU (Advanced Micro Devices, Inc., Santa Clara, CA, USA) and an NVIDIA GeForce RTX 4090 GPU (NVIDIA Corporation, Santa Clara, CA, USA) with 24 GB of VRAM.

5 Discussion: Limitations and Future Research

The results show that the integrated framework is most useful for separating physical mobility-ratio effects from numerical connectivity effects in high-resolution fingering simulations. The 1D verification confirms that the fractional-flow and relative-permeability components reproduce the expected saturation-transport behavior, while the Habermann benchmark shows that the same formulation captures the primary mobility-ratio dependence of sweep efficiency and fingering morphology. The stencil comparisons further demonstrate that connectivity is not a secondary numerical detail; it can alter breakthrough behavior and the apparent three-dimensional structure of fingers. Thus, the principal implication of this study is that reservoir-relevant saturation-dependent mobility and grid-connectivity control must be considered together when interpreting high-resolution 3D viscous fingering simulations.

The present framework was intentionally evaluated under viscosity-dominated, homogeneous, and Newtonian displacement conditions so that the effects of mobility contrast, saturation-dependent mobility, and stencil connectivity could be isolated. This controlled setting allowed the simulator to reproduce the main mobility-ratio

dependence observed in the Habermann benchmark and to demonstrate how three-dimensional fingering morphology is affected by stencil connectivity. However, this also defines the main directions for future work.

First, quantitative three-dimensional validation should be performed using raw X-ray CT or micro-CT saturation fields. Such datasets would allow direct comparison of invaded volume, interfacial area, connected-cluster count, dominant-channel volume fraction, and breakthrough evolution. These metrics would provide a stronger basis for distinguishing physical finger growth from numerical connectivity effects.

Second, the physical model should be extended beyond the viscosity-dominated homogeneous cases considered here. Incorporating capillary pressure, gravity segregation, permeability heterogeneity, anisotropy, and more realistic inlet/outlet conditions would allow the framework to examine transitions among viscous fingering, capillary fingering, gravity override, and heterogeneity-controlled channeling. This extension is particularly important for applying the method to reservoir and geological-storage settings.

Third, the constitutive description should be generalized to more complex displacement systems, including polymer flooding, non-Newtonian shear-dependent viscosity, and CO_2 -brine displacement with density and compressibility contrasts. These extensions would test whether the integrated relative-permeability, fractional-flow, CFD, stencil-connectivity, and GPU-solver framework remains robust when additional physical mechanisms are introduced.

Finally, systematic numerical studies are needed to quantify grid-refinement behavior, stencil-weight sensitivity, and GPU scalability. These analyses would clarify the tradeoff between reduced grid-aligned propagation and increased computational cost, and would provide guidance for transferring resolved three-dimensional fingering descriptors into larger-scale reservoir simulation workflows.

6 Conclusion

This study developed a high-resolution three-dimensional CFD framework for viscosity-dominated fingering in porous media. The main contribution is the integration of reservoir-style saturation-dependent mobility with direct three-dimensional resolution of fingering morphology. Specifically, the simulator couples a pressure-based porous-media flow formulation with Corey-type relative permeability, fractional-flow phase partitioning, residual saturation constraints, D3Q27-type stencil connectivity, and a GPU-accelerated AMG-GMRES solver. This combination addresses the methodological gap identified in the Introduction: relative permeability, fractional flow, CFD, isotropic stencil connectivity, and GPU-based sparse linear solution have each been used in related contexts, but their unified use for resolving three-dimensional fingering with reservoir-relevant mobility has not been systematically established.

First, the one-dimensional Buckley-Leverett verification confirmed that the implemented fractional-flow and Corey-type relative permeability formulations reproduce the expected analytical displacement behavior. This result verifies the fundamental saturation-transport component of the framework before applying it to multidimensional fingering problems.

Second, the quasi-two-dimensional quarter five-spot simulations reproduced the main mobility-ratio-dependent behavior reported by Habermann. Stable cases produced piston-like displacement and high sweep efficiency, whereas unstable cases developed shielding, branching, and dendritic fingering. The sweep-efficiency comparison also captured the experimentally observed decreasing trend with increasing mobility ratio. Differences in exact finger topology and recovery magnitude are expected because the ideal homogeneous simulations do not include the pore-scale heterogeneity and transverse dispersion present in physical sand-pack experiments.

Third, the stencil-connectivity analysis showed that numerical connectivity materially affects the computed fingering morphology. In quasi-two-dimensional tests, the D3Q27-type stencil reduced preferential grid-aligned channeling relative to the conventional 7-point stencil. In the three-dimensional $M = 71.5$ case, the D3Q27-type stencil increased breakthrough PVI from 0.05 to 0.22 and produced broader radial spreading, although this improvement was accompanied by higher per-iteration computational cost. This comparison supports the use of D3Q27-type connectivity for the subsequent three-dimensional simulations, while it should be interpreted as a numerical sensitivity analysis rather than a complete morphological validation.

Finally, the three-dimensional simulations revealed volumetric fingering behavior that is inaccessible to quasi-two-dimensional models. The invading phase formed interconnected tubular channels, bypassed resident fluid in the depth direction, and distorted the pressure field around dominant finger tips. These results indicate that pressure shielding and peripheral bypass are inherently three-dimensional features of unstable displacement under high mobility ratio.

Overall, the proposed framework provides a mass-conservative solution and computationally feasible basis for studying three-dimensional viscous fingering under viscosity-dominated conditions. Future work should extend the model to include gravity segregation, capillary effects, permeability heterogeneity, and CO_2 -brine displacement, and should validate the predicted three-dimensional morphology using raw saturation fields from X-ray CT or other volumetric imaging data.

7 Acknowledgments

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Statements & Declarations

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Competing Interests

The authors have no relevant financial or non-financial interests to disclose.

Author Contributions

All authors contributed to the conception and design of the study. Investigation, material preparation, and formal analysis were performed by Chenxi Li and Yanbing Tang. Chenxi Li wrote the first draft of the manuscript. Software development and visualization were carried out by Chenxi Li and Yanbing Tang. Resources, project administration, supervision, review and editing were provided by Hisanao Ouchi and Masanori Kurihara. All authors commented on previous versions of the manuscript and approved the final version.

Data Availability

The datasets generated and analyzed during the current study are available from the corresponding author on reasonable request. The custom simulation code is not publicly available because it is part of an in-house solver under ongoing development.

Supplementary Files

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